Bengaluru residential prices

What actually drives the number on the board, and how well it can be predicted

Data science case study | 13,320 listings | All figures in Indian rupees (Rs), prices in lakh

Summary

A gradient-boosted model prices a Bengaluru listing to a median error of 17.7%, close enough to flag a quote that is badly out of line. It beats a naive median-price guess by Rs 28 lakh per listing (Rs 27.6 lakh average error against Rs 56.0 lakh). The two features that do the work are unglamorous: built-up area first, locality second.

11,423
LISTINGS ANALYSED
from 13,320 raw rows, 14.2% discarded

4.8x
LOCALITY SPREAD
Rs 2,885 to Rs 13,719 per sq ft

Rs 27.6 L
AVERAGE ERROR
held-out listings, R-squared 0.68

10%
BEDROOM PREMIUM
extra cost a 3BHK carries beyond its floor area

Four findings worth acting on

- **Location is worth 4.8x.** The same square foot costs Rs 2,885 in Attibele and Rs 13,719 in Rajaji Nagar. No other variable in the data spans that range.
- **A 3BHK is priced as more than the extra floor.** Across 156 localities, a 3BHK asks 51.3% more than a 2BHK while offering 37.0% more area — about 10.5% of the jump buys configuration, not square feet.
- **Area type is overrated.** Carpet area carries only a 14% premium per square foot over super built-up, roughly the loading factor itself.
- **The model fails at the top.** Above Rs 150 lakh it under-predicts systematically; tree ensembles cannot extrapolate into a sparse luxury tail.

One row in seven could not be trusted

The source file holds 13,320 listings across 9 columns. Two problems dominate: **society** is 41.3% missing, and **total area** is stored as text — ranges like “1000 - 1500”, plus areas given in square metres, yards, acres, cents, guntha, perch and grounds. Both are handled explicitly rather than dropped silently.

Cleaning applied five rules, each counted below: parse area to square feet, keep physically plausible area per bedroom (at least 300 sq ft), cap bathrooms at two more than bedrooms, trim price per square foot beyond three standard deviations of its own locality, and drop exact duplicates. The result is a sample of 11,423 listings.

Cleaning stage	Rows kept	Removed
raw rows	13,320	0
unparseable area/config/price	13,304	16
area outside 300-10,000 sq ft	13,251	53
BHK outside 1-10	13,239	12
bathroom count missing	13,182	57
more bathrooms than rooms allow	13,167	15

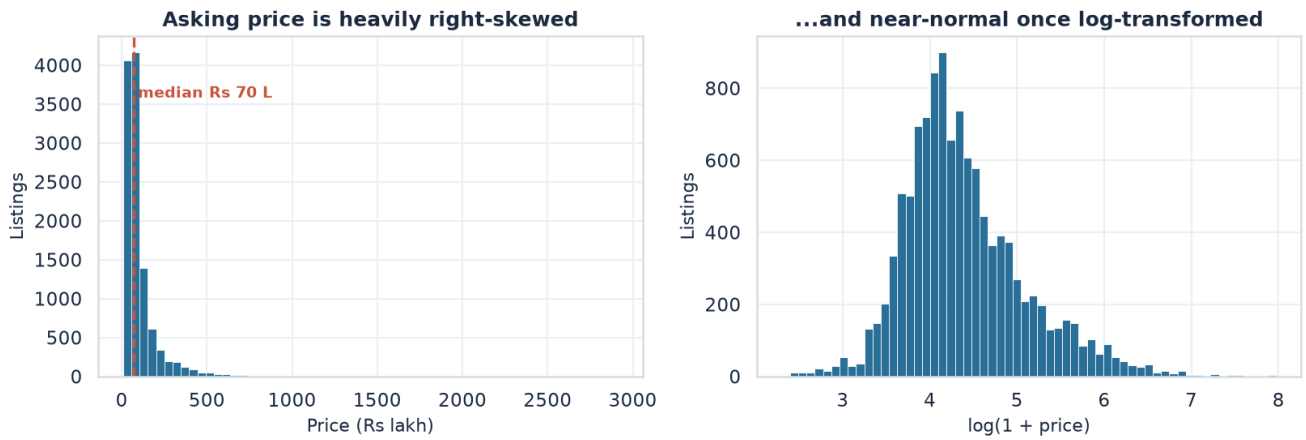
Cleaning stage	Rows kept	Removed
under 300 sq ft per bedroom	12,446	721
price/sq ft beyond 3 sd of locality	12,298	148
exact duplicates	11,423	875

Locality strings were the last problem: 1,305 distinct spellings for what is a few hundred neighbourhoods. Localities with fewer than 10 listings were folded into a single group, leaving 227 usable categories. The cleaned sample has a median asking price of Rs 70 lakh for 1,305 sq ft — Rs 5,311 per square foot.

Area sets the price, locality sets the rate

Asking prices are heavily right-skewed, with a long tail of luxury listings that would dominate any untransformed model. Working in log-price makes the distribution near-normal and is the single most useful modelling decision here.

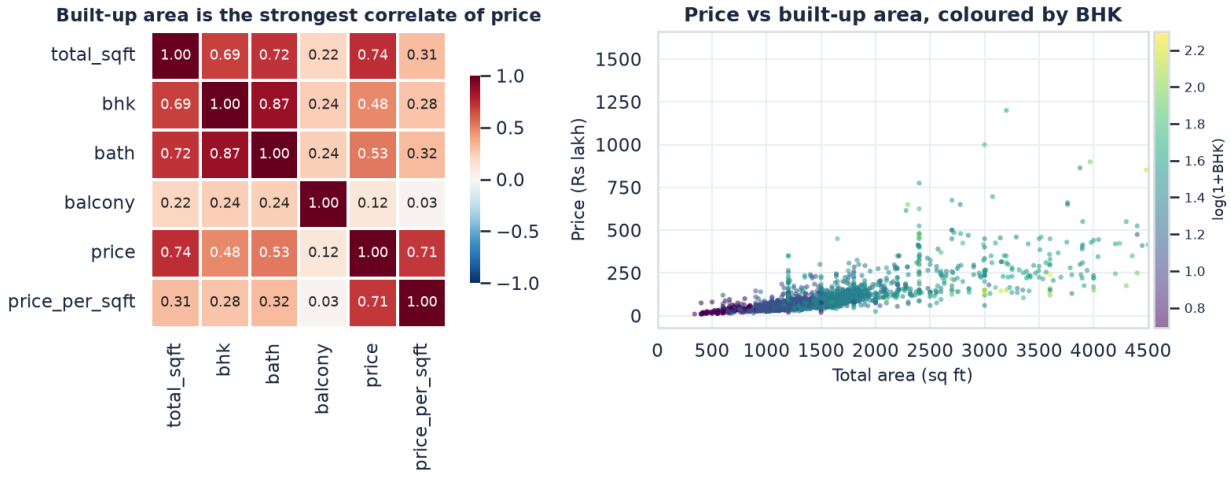
Figure 1 - Distribution of residential asking prices



Source: cleaned sample, n = 11,423 listings.

Built-up area correlates with price at 0.74, stronger than any other variable. Bathrooms and bedrooms correlate at 0.87 with each other, meaning they carry nearly the same information — worth remembering when reading feature importance later.

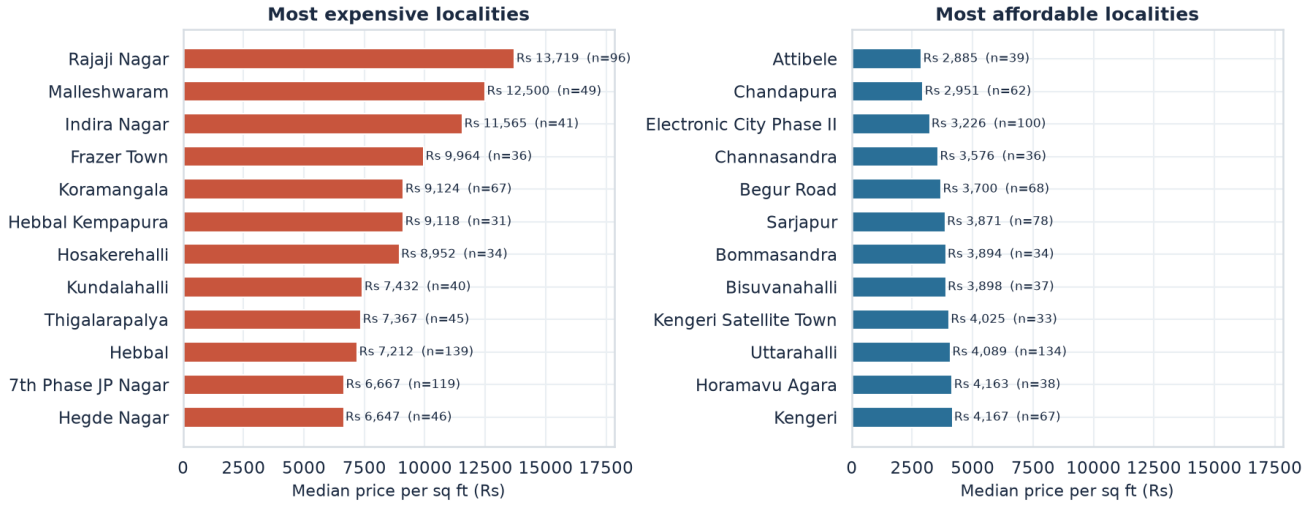
Figure 4 - What moves price, and what moves with it



Pearson correlations on the cleaned sample; scatter shows a 2,500-listing sample.

The locality spread is the finding a buyer should remember: Rs 2,885 per square foot in Attibele against Rs 13,719 in Rajaji Nagar — a 4.8x difference within one city.

Figure 3 - The same square foot costs nearly 5x more across the city

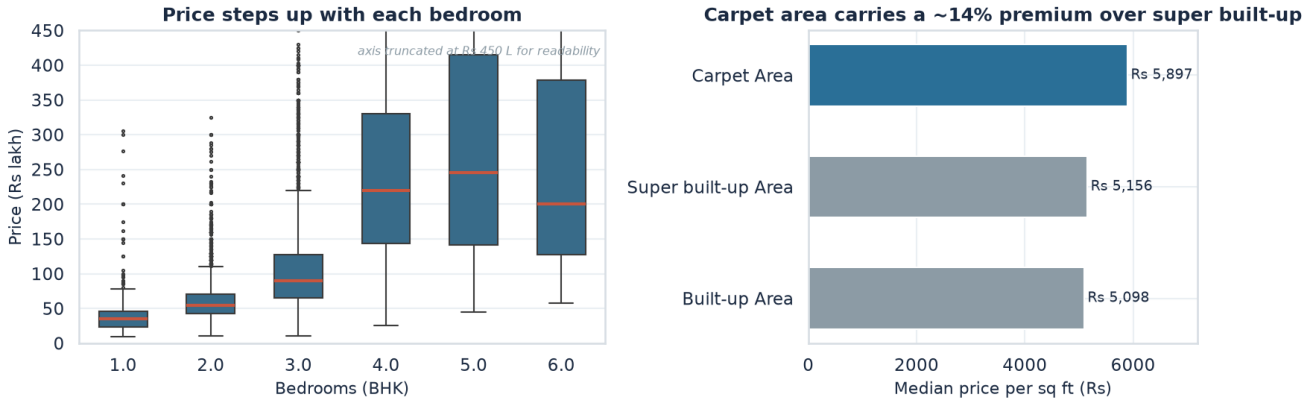


Localities with at least 30 listings; n shown beside each bar.

An extra bedroom costs 51% more for 37% more floor

Comparing 2BHK and 3BHK listings within the same locality — 156 localities had at least five of each — the median 3BHK asks 51.3% more while offering 37.0% more built-up area. The gap of about 10.5% is what the configuration itself costs, separate from the square feet. A buyer comparing a 2BHK and a 3BHK in the same building should expect to pay for more than the extra room.

Figure 2 - Price by configuration and area type



Left: price distribution by configuration, axis truncated at Rs 450 lakh. Right: median rate per square foot by area type; plot area excluded as land is priced on a different basis.

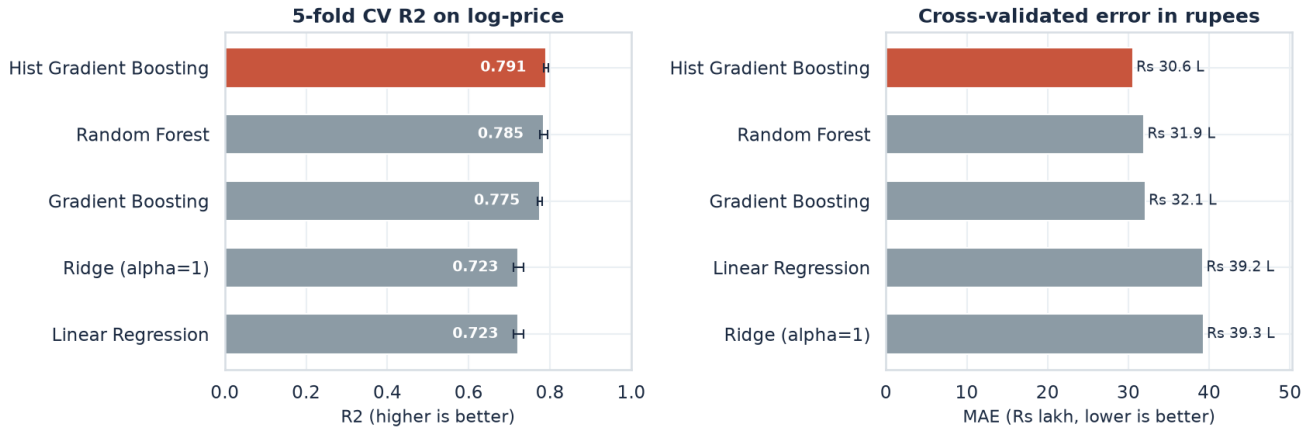
Gradient boosting beats the linear baseline by Rs 9 lakh a listing

The target is log price. Features are built-up area, bedrooms, bathrooms, balconies, area per bedroom, a ready-to-move flag, area type and locality group. All preprocessing — median imputation, scaling, one-hot encoding with rare categories folded — sits inside a pipeline, so nothing leaks across folds. Models were compared on 5-fold cross-validation over 9,138 training listings; the 2,285 held-out listings were touched once, at the end.

Model	CV R-squared (log)	MAE	RMSE
Hist Gradient Boosting	0.791	Rs 30.6 L	Rs 82.7 L
Random Forest	0.785	Rs 31.9 L	Rs 94.6 L
Gradient Boosting	0.775	Rs 32.1 L	Rs 86.3 L
Linear Regression	0.723	Rs 39.2 L	Rs 158.6 L
Ridge (alpha=1)	0.723	Rs 39.3 L	Rs 160.3 L

5-fold cross-validation on the training split. Errors are in rupees lakh on the original price scale, after back-transforming from log.

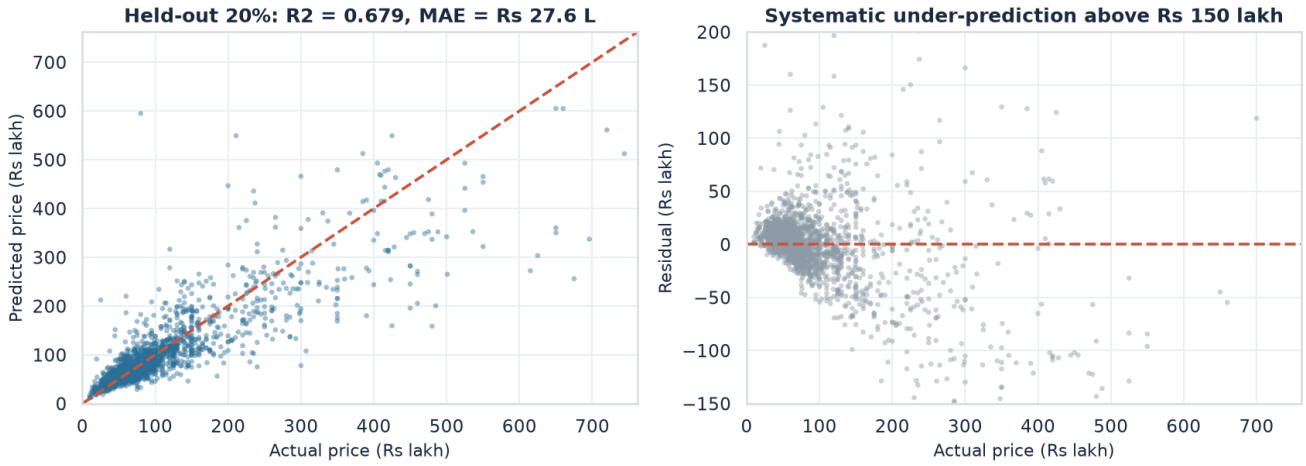
Figure 5 - Tree ensembles beat the linear baselines by a wide margin



Error bars show standard deviation across the five folds.

On the held-out fifth of the data the chosen model scores R-squared 0.80 in log space and 0.68 on raw price, with an average error of Rs 27.6 lakh. Half of all listings land within 17.7% of their actual price; 55.3% land within 20% and 72.6% within 30%. For context, guessing the median price every time is off by Rs 56.0 lakh on average.

Figure 6 - Where the model is right, and where it isn't



Held-out set, n = 2,285 listings. Residual is predicted minus actual price.

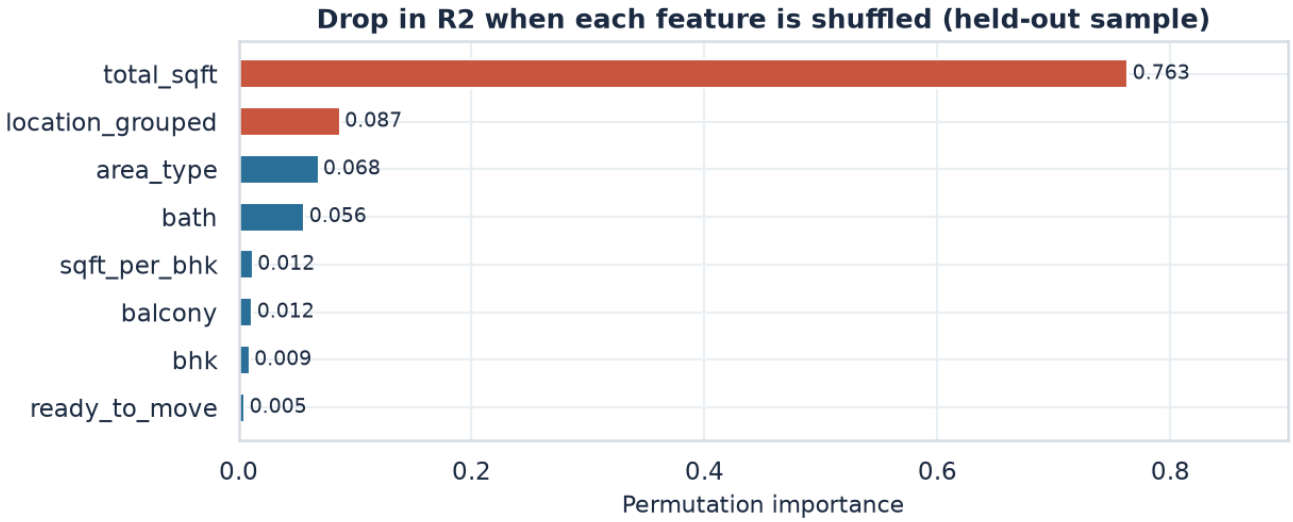
The model under-prices the top of the market

Below Rs 150 lakh the residuals are symmetric — the model is unbiased. Above it, residuals skew negative, meaning expensive properties are consistently under-predicted. This is structural rather than a tuning problem: tree ensembles cannot extrapolate beyond the price range they saw in training, and luxury listings are both rare and heterogeneous. Any production use would need a separate treatment for the top end, or an explicit confidence band rather than a point estimate.

Area dominates, and the importance chart understates location

Permutation importance on the held-out set puts built-up area far ahead of everything else. Locality ranks second at 0.087 — but that figure understates its real role, because one-hot encoding spreads location across roughly 140 columns, so no single column looks important. Read importance per group, not per column. Bedrooms rank last because area and bathrooms already carry the same signal.

Figure 7 - Size first, location second, everything else marginal



Drop in R-squared when one feature is shuffled, measured on a 500-listing held-out sample.

What to do with this

- **Use it as a sanity check, not a valuation.** A quote more than 25% away from the model's estimate is worth a second look at comparable listings in the same locality.
- **Price by locality first, then by configuration.** Marketing a 3BHK on floor area alone ignores the ~10% configuration premium the market clearly charges.
- **Do not deploy on the luxury segment.** Above Rs 150 lakh the model under-predicts; route those listings to manual comparables until more data exists.

Limitations

- **Asking prices, not transactions.** Listed prices in India typically sit above the eventual deal; the model estimates the quote, not the cheque.
- **One snapshot, no time dimension.** Availability dates cluster in 2018, so the data cannot capture the market cycle — a serious gap for present-day valuation.
- **No builder or amenity signal.** Society is 41% missing, removing the strongest quality variable a human valuer would use.

- **Locality labels are merged.** 1,305 raw spellings collapsed to 227 groups; some distinct neighbourhoods now sit in the Other bucket.

Next steps

- Add listing dates to model price trend, not just price level.
- Bring in external features: distance to metro and tech parks, school density, floor number, building age.
- Switch to quantile regression to output a price range — closer to what a buyer needs.
- Validate against registered transaction data (Kaveri/IGR) rather than portal listings.

Reproducing this analysis

Everything in this report is generated from one notebook. The notebook cleans the raw file, produces all seven figures, compares the five models and writes the metrics file this document reads from — so no number here was typed by hand.

File	Contents
bengaluru_housing_analysis.ipynb	Full analysis, executed, with outputs
bengaluru_house.csv	Raw source data, 13,320 rows
metrics.json	Every figure quoted in this report
figures/	Seven charts, 150 dpi

Runtime is about three minutes on a single CPU. Random seed 42 throughout; scikit-learn 1.9, pandas 3.1. Source data is the public Bengaluru house-price listings dataset, retrieved from a GitHub mirror of the original Kaggle release.